EXECUTIVE SUMMARY

for

Proposed Stone Khanivade Stone Quarry Project

at

Gat No. 9 Part, Lease Area = 15.39 Ha.

Village: Khanivade, Taluka: Palghar, District: Palghar, Maharashtra

Type of Land: Reserve Forest Land

Proposed By

JAWAHARLAL NEHRU PORT AUTHORITY (JNPA)

Consultant

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ABBREVIATION

Sr. No	Abbreviation	Full Form
1	JNPA	Jawaharlal Nehru Port Authority
2	MMB	Maharashtra Maritime Board
3	VPPL	Vadhvan Port Project Limited
4	EIA	Environmental Impact Assessment
5	EMP	Environmental Management Plan
6	PMCP	Progressive Mine Closure Plan
7	DGM	Directorate of Geology and Mining
8	MoEFCC	Ministry of Environment, Forest and Climate Change
9	MPCB	Maharashtra Pollution Control Board
10	CRZ	Coastal Regulation Zone
11	DTEPA	Dahanu Taluka Environment Protection Authority
12	IBM	Indian Bureau of Mines
13	NH	National Highway
14	MSEB	Maharashtra State Electricity Board
15	ROM	Run of Mine
16	TPA	Tonnes Per Annum
17	MMT	Million Metric Tonnes
18	RL	Reduced Level
19	CPCB	Central Pollution Control Board
20	SPM	Suspended Particulate Matter
21	PPE	Personal Protective Equipment
22	IMEEC	India-Middle East-Europe Economic Corridor
23	MMME Rules	Maharashtra Minor Mineral Extraction Rules
24	MCDR	Mineral Conservation and Development Rules

1. INTRODUCTION

Jawaharlal Nehru Port Authority (JNPA[1]), is a major port under Ministry of Shipping and Waterways Government of India is developing Vadhvan Port in the Palghar district of Maharashtra in collaboration with the Maharashtra Maritime Board. Designed as a "greenfield" project, it will be a deep-water, all-weather international port managed by Vadhvan Port Project Limited (VPPL), a Special Purpose Vehicle where JNPA holds a 74% stake and the Maharashtra Maritime Board holds 26%. This project is being implemented under the central government's Sagarmala Programme, and its foundation stone was laid by the Prime Minister on August 30, 2024.

The port requires over 64.5 Million Metric Tonnes (MMT) of basalt armour rock and aggregate for construction of a 10.14 km long breakwater. JNPA has identified private stone quarry sites roughly 20 km from the port to meet these material needs, specifically proposing mining on 15.39 hectares of forest land at Gargaon Stone Quarry Panch-5. The Maharashtra State Forest Department has recommended this proposal to the Ministry of Environment, Forest and Climate Change (MoEFCC) to approve the diversion of this forest land for the project's use.

This Executive Summary presents, in simple and clear language, the key aspects of the Environmental Impact Assessment (EIA[2]) Report prepared for the proposed Khanivade Stone Quarry Patch-5. It covers the project description, environmental baseline, anticipated impacts, mitigation measures, forest clearance status, compensatory afforestation, and environmental management plan.

Table 1 Project at a glance

Project Name	**Khanivade Stone Quarry — Patch 1**
Project Proponent	Jawaharlal Nehru Port Authority (JNPA)
Location	Gat No. 9 Part, Village Khanivade, Taluka & Dist. Palghar, Maharashtra
Lease Area	15.39 Ha (Reserve Forest Land)

[1] JNPA : Jawaharlal Nehru Port Authority
[2] EIA : Environmental Imapact Assesment

Project Name	Khanivade Stone Quarry — Patch 1
Mineral	Basalt (Minor Mineral) — hard rock, specific gravity 2.90
End Use	Captive supply for Vadhvan Port construction (breakwater, shore protection, aggregate)
Distance to Vadhvan Port	17.2 km (north-west of lease centroid)
EIA Category	Category B1 — EIA Notification, 2006 (open-cast, RF land, 5-25 Ha); Appraised by SEIAA Maharashtra
Annual ROM[3] Production	25,85,553 Tonnes per Annum (TPA[4])
Annual Saleable Production	23,26,998 TPA (90% yield from ROM)
Total Mineable Reserves	1,29,27,769 Tonnes (Proved)

2. PURPOSE OF STUDY

The primary purpose of the mine is to provide an internal supply of basalt stones (boulders), armor stones, and crushed grit for the Vadhavan Port project. In the absence of a nearby internal source, stones would have to be brought from mines located 80–120 km away from the port. This would result in a negative impact on the environment and increase transportation costs by approximately 200–300 rupees per ton. Additionally, diesel consumption and truck traffic would increase fivefold, creating a risk of delay for this project of national importance.

Table 2 Clearances Required and status

Sr.	Clearance / Approval	Authority	Status
1	Environmental Clearance (EC)	SEIAA[5] Maharashtra (B1 Category)	Applied — this Environmental Impact Assessment

[3] ROM : Removal of Mine
[4] TPA : Trillion Per Annum
[5] SEIAA : State Environmental Impact Assessment Authority

Sr.	Clearance / Approval	Authority	Status
			(EIA) for Public Hearing
2	Forest Clearance Stage-I	MoEFCC[6] / GoM	Is in process
3	Forest Clearance Stage-II	MoEFCC, New Delhi	In Progress
4	Mining Plan Approval	DGM, Government of Maharashtra	Received — 09.02.2026
5	Provisional Letter of Intent (LoI)	District Collector, Palghar	Received — 20.01.2026
6	Grampanchayat NOC	Grampanchayat Khanivade-Khanivade	Received — 03.02.2026
7	FRA 2006 Compliance	Sub-Divisional Officer, Palghar	FRA (Forest Rights Act) proposals have been submitted to the Gram Panchayat for consideration.
8	MPCB[7] Consent to Establish	Maharashtra Pollution Control Board	To be applied after EC

3. EXTENT OF STUDY AND STUDY AREA

The mining lease area for Khanivade is located at Gat No. 9 Part, Reserve Forest Land, Village Khanivade, Taluka Palghar, District Palghar, Maharashtra, with an approved lease area of 15.39 Ha

The study covers all aspects of the lease area including geology, reserves estimation, mining methodology, blasting parameters, drainage, waste disposal, environmental management, and progressive mine closure planning. Baseline environmental data covering air, water, noise, soil, ground water, surface water land use, flora, and fauna have been collected and analysed for the lease area and surroundings for the period of 3 month (December 2025-February 2026). The study area for the proposed project covers a radial distance of 10 km from the project site, delineated with reference to Survey of India Toposheet No. E43A/13.

Assessment of Impact due to Blast

As compliance to terms of reference dated 25/04/2026, the study area has been examine viz-a-viz the MPCB Circular dated 13/08/2020 regarding siting criteria for stone quarries in the state of Maharashtra, the regulations for Danger zone is 500 m prescribed by the Directorate General of Mines Safety. There are seven villages falling within the 500-meter danger zone of the quarry mining project area. These villages are Gargaon, Khaniwade, Ravate, Sumdi, Mahagaon, Chinchare, and Valve.



Figure 1.1 Google Image

4. GEOLOGICAL RESERVES AND MINE PLAN

The lease is entirely underlain by Deccan Trap Basalt (Upper Cretaceous — Lower Eocene age). Reserves have been estimated by four geological cross-sections (A-A' to D-D') using the Cross-Section Method with a tonnage factor of 2.9 (specific gravity of basalt).

4.1 Reserve Summary

Table 3 Reserve Summary

Total Geological Reserves (Proved)	1,43,64,187 Tonnes
Less: Ore Blocked in Pit Slopes (10%)	14,36,418 Tonnes

Total Mineable Reserves	1,29,27,769 Tonnes
Annual ROM Production	25,85,553 TPA
Annual Saleable (@ 90% yield)	23,26,998 TPA
Mine Life	5 Years

4.2 Year-wise Production and Overburden

Table 4 Year-wise Production and Overburden

Year	OB Generated (T)	ROM Ore (T)	Saleable Ore @ 90% (T)	Rejects @ 10% (T)
Year 1	2,14,565	25,85,553	23,26,998	2,58,555
Year 2	1,19,271	25,85,553	23,26,998	2,58,555
Year 3	1,22,960	25,85,553	23,26,998	2,58,555
Year 4	2,26,246	25,85,553	23,26,998	2,58,555
Year 5	2,34,238	25,85,553	23,26,998	2,58,555
TOTAL	9,17,280	1,29,27,765	1,16,34,990	12,92,777

4.3 Mining Method and Equipment

Mining is by open-cast method, working top-down in benches of 6 m height x 6 m width with an overall pit slope of 45°. Blast holes of 34/100 mm diameter and 1.5–1.6 m depth are drilled using jack-hammer drills. Blasting uses gelatin (500 g/hole, maximum 20 holes/round, charge per delay limited to ≤5 kg) with NONEL (shock-tube) initiation — a safe, non-electrical system that avoids mis-fires. Monthly explosives limit: 300 kg gelatin + 700 detonators. Blasting is restricted to 10:00 hrs to 17:00 hrs only.

Table 5 Machinery Required

Equipment Type	Qty.
450 CFM Crawler Drill Machine	3
Jack Hammer with Tractor Compressor	4
Excavator 220	6
Excavator 380	3
Excavator 450	1

Equipment Type	Qty.
Dumper 10 Wheeler	25
62.5 KVA DG Set)	2
Water Tanker 12000 Ltr	2
Dewatering Pump 20 HP	2
Dozer D-80	1

In this project, controlled blasting with 100 mm diameter is proposed.

5. RESOURCE REQUIREMENTS

5.1 Water

There will be no water requirement during construction phase. 0.45 kld water will be required during operational phase for domestic use and some operational activities.

5.2 Manpower

Table 6 Manpower Requirement

Category	Number
Mine Foreman (Diploma in Mining Engineering)	1
Mine Mate / Licensed Blaster	1
Supervisor / Clerk	1
Skilled Workers	4
Unskilled Workers	2
TOTAL DIRECT	9
Indirect (transport, security, canteen, plantation)	25–30

6. FOREST LAND, TREE FELLING AND COMPENSATORY AFFORESTATION

The entire lease area of 15.39 Ha is Reserve Forest land. Mining requires formal diversion of this land under the Forest (Conservation) Act, 1980. The following has been done by JNPA.

6.1 Forest Clearance Status

Stage-I Forest Clearance is in process vide Proposal No. FP/MH/MIN/QRY/550969/2025. The proposal covers 90.9483 Ha of forest land for the entire Vadhvan Port quarry and road/railway cluster (quarry patches + approach roads + road widening). Key authorities that have recommended the proposal:

- Deputy Conservator of Forests, Dahanu Forest Division — recommended
- Chief Conservator of Forests (T), Thane — recommended
- Addl. Principal Chief Conservator of Forests and Nodal Officer, Nagpur — forwarded to State Government
- Principal Chief Conservator of Forests (HoFF), Maharashtra State, Nagpur — recommended for acceptance
- District Collector, Palghar — confirmed national importance, recommended

Stage-II Forest Clearance from MoEFCC, New Delhi is being processed. Mining will commence only after Stage-II clearance is received.

6.2 Net Present Value (NPV[8])

NPV of the forest area to be diverted has been calculated by DCF Dahanu as Rs. 14,51,34,388/- (Rupees Fourteen Crore Fifty-One Lakh Thirty-Four Thousand Three Hundred and Eighty-Eight). JNPA has furnished a formal undertaking to pay this full amount to the CAMPA fund before commencement of any forest land use.

6.3 Tree Felling

A detailed tree enumeration survey has been carried out by the Forest Department across the entire diverted area. Total trees in the 90.9483 Ha diverted area: 45,554. Trees falling within Khanivade Stone Quarry Patch-5 and its associated corridors: 15,116 trees. The tree species commonly found include Karanja (Pongamia pinnata), Savar (Bombax ceiba), Ain/Behada (Terminalia bellirica), Teak (Tectona grandis), Palas (Butea monosperma), Wad (Ficus benghalensis), and Jambhul (Syzygium cumini).

[8] NPV : Net Present Value

Table 7 Girth-wise count of all trees in the diverted area (DCF Dahanu certified):

Girth Class (cm)	0–30	31–45	46–60	61–75	76–90	91–105	106–120	121–135	136–150	Above 151	TOTAL
Number of Trees	760	7,387	6,952	5,957	5,693	5,304	7,227	1,938	2,792	1,544	45,554

Tree felling will be carried out only after receipt of Stage-II Forest Clearance from MoEFCC and under the supervision of the Range Forest Officer, Boisar. All trees will be felled as per the approved list only.

6.4 Compensatory Afforestation (CA[9]) — Parbhani District

The Divisional Forest Officer (DFO), Hingoli has identified and certified 100.19 Ha of non-forest land in Parbhani District as suitable for Compensatory Afforestation — free from encroachment. This land exceeds the total forest area diverted (90.9483 Ha), meeting the legal requirement:

Table 8 Compensatory Afforestation:

Sr.	District	Tehsil	Village	Gut No.	CA Area (Ha)
1	Parbhani	Gangakhed	Supa Tanda	132	10.00
2	Parbhani	Jintur	Mandhani	62	23.58
3	Parbhani	Jintur	Akoli	147	11.07
4	Parbhani	Jintur	Saykheda	107	17.70
5	Parbhani	Jintur	Jamb Khurd	186	11.44
6	Parbhani	Jintur	Raykheda	154	26.40
			TOTAL		100.19 Ha

CA scheme cost: Rs. 9,00,59,781/- for 93 Ha plantation (75 Ha fresh plantation for 10 years + 18 Ha maintenance of existing plantation). JNPA has furnished a formal undertaking to pay the full CA cost to MAHA CAMPA, Nagpur. Geo-

[9] CA : Compensatory Afforestation

referenced maps, village maps, and DFO[10] Suitability Certificates for all six CA parcels are submitted with the Forest Clearance proposal.

6.5 Plantation

PP has to provide 15,390 saplings within the leas covering 6.156 Ha — 40% of the total lease area — at a density of 2,500 saplings per Ha. As Stone Quarry is located in Forest Clearance where there is already flora, plantation will be done at Parbhani allotted by Forest Department.

Table 9: Plantation on 6.156 Ha

Location	Area (Ha)	Saplings
Peripheral safety zone (greenbelt)	1.34	3,350
OB dump top and slopes	1.66	4,150
Reclaimed benches (Year 3 onwards)	1.85	4,625
Internal haul road sides	0.74	1,850
Site office / facility area	0.40	1,000
Topsoil pad and miscellaneous	0.166	415
TOTAL	6.156 Ha	15,390 saplings

Species: Karanja, Teak, Behada, Palas, Neem, Wad, Peepal, Mango, Jambhul, Savar, Kanchan, Amaltas, Chinch — all native. Survival targets: ≥70% at Year-1; ≥85% at Year-3.

7. ENVIRONMENTAL BASELINE (EXISTING CONDITIONS)

Primary baseline environmental monitoring was carried out by an NABL-accredited laboratory for the winter season (December 2025 — February 2026) at the project site and in a 5–10 km buffer zone.

[10] DFO : District Forest Officer

7.1 Physical Setting

Table 10 Physical Setting:

Geology	Deccan Trap Basalt (Upper Cretaceous —Lower Eocene); compact, hard, SG 2.90
Topography	Gently undulating basaltic plateau; highest point in the area is 123.979 m RL.; slope 2–5%
Annual Rainfall	2,500 mm (June–September monsoon)
Temperature	Maximum 36°C (May); Minimum 18°C (January)
Drainage	Dendritic seasonal drainage; no perennial streams within lease; Vaitarna river 3.2 km NW
Land Use (lease)	Entirely degraded scrub Reserve Forest; no agriculture, no habitation
Soil	Shallow basaltic loamy soil; pH 6.7–7.1; no heavy metal contamination

7.2 Ambient Air Quality (Baseline — Winter Season 2025-26)

Five AAQ[11] monitoring stations were established. All baseline values are well within NAAQS[12] 2009 limits:

Table 11 AQI Baseline Data:

Station	Location	PM10 (µg/m³)	PM2.5 (µg/m³)	SO2 (µg/m³)	NOx (µg/m³)	NAAQS PM10 Limit
AAQ-1	Project Site, Khanivade Village	47	16	6	10	100
AAQ-2	Lalonde Talathi Office	52	21	6	12	100
AAQ-3	Tata Housing Road, Boisar	63	28	16	23	100
AAQ-4	Profectuals School Gate	57	22	7	14	100
AAQ-5	Gargaon Village	50	21	6	12	100

[11] AAQ : Ambient Ait Quality
[12] NAAQS : National Ambient Ait Quality Standard

7.3 Ground Water Quality (Baseline)

9 monitoring wells (GW-1 to GW-9) were sampled during pre-monsoon (April 2026). Results are generally within IS 10500:2012 (Drinking Water Standard). Parameters monitored: pH, TDS[13], hardness, chloride, sulphate, nitrate, fluoride, iron, total coliform, oil and grease.

Table 12 Ground Water Baseline Data:

Station	Location	pH	TDS (mg/L)	Remarks
GW-1	Project Site	7.3	210	Within IS 10500:2012
GW-2	Nandini Charitable Trust	7.5	468	Within IS 10500:2012
GW-3	Ravate Village	7.8	414	Within IS 10500:2012
GW-4	Lolande Village	7.3	377	Within IS 10500:2012
GW-5	Tata Housing Society	7.6	478	Within IS 10500:2012
GW-6	Dream City Society	7.6	395	Within IS 10500:2012
GW-7	ZP School Ambat pada	7.6	458	Within IS 10500:2012
GW-8	BMT English High School	7.5	454	Within IS 10500:2012

7.4 Ambient Noise (Baseline)

Table 13 Ambient Noise Baseline Data:

Location	Leq Day dB(A)	Leq Night dB(A)	Applicable Standard Day	Applicable Standard Night
Within Lease (Industrial zone)	-	-	75 dB(A)	70 dB(A)
Khanivade Village (Residential)	52.0	42.4	55 dB(A)	45 dB(A)
Wave Village (Residential)	52.5	42.5	55 dB(A)	45 dB(A)
PHC / School Zone (Silence zone)	49.5	39.5	50 dB(A)	40 dB(A)

[13] TDS : Total Dissolve Solids

7.5 Flora and Fauna

- The lease is classified as degraded scrub Reserve Forest (Champion & Seth Type 5C — Southern Tropical Dry Deciduous Scrub).
- Vegetation is sparse — stunted shrubs, grasses and rocky outcrops with canopy cover less than 30%.
- No Schedule-VI plant species (WPA 1972) and no IUCN Red-Listed plant species were recorded.
- Common fauna: Indian hare, Indian fox, common monitor lizard (Schedule I — WPA), peacock (Schedule I — WPA), common krait, Indian roller, black kite and various Schedule-IV birds.
- No Schedule-I megafauna (tiger, leopard, elephant) found anywhere in the 10 km study area.
- No Protected Area, ESZ, Tiger Reserve, Elephant Reserve, or Ramsar site lies within 10 km. Nearest PA: Tansa Wildlife Sanctuary is more than 18 km south-east.

7.6 Socio-Economic Profile

Table 14 Socio-Economic Data:

Host Village	Khanivade — Population 966; Literacy 53.40%; ST population 948; BPL 28%
Habitation within lease	NONE — lease is uninhabited Reserve Forest land
Nearest habitation	Khanivade village (0.80 km from lease boundary)
Primary occupation	Agriculture, daily wage labour
Key infrastructure	Adhikari Lifeline Hospital (3.30 km), ZP School in Khanivade village, Boisar Railway Station (6 km)
Drinking water	Open dug-wells and tankers; no piped supply in Khanivade

8. ANTICIPATED ENVIRONMENTAL IMPACTS AND MITIGATION MEASURES

8.1 Air Environment

The main air quality concern is fugitive dust from drilling, blasting, haul roads and material handling. AERMOD air dispersion modelling predicts:

- Maximum incremental PM10 at nearest receptor (Khanivade South, 0.8 km):

5.2 µg/m³

- Resultant PM10 at Khanivade with project: 47 µg/m³ — well within NAAQS 2009 limit of 100 µg/m³
- Incremental PM2.5 at Khanivade: 2.5 µg/m³ — within NAAQS limit of 60 µg/m³

Mitigation: Wet drilling on all drill rigs at all times; water tanker watering of haul roads 4 rounds per shift; chemical surfactant in summer months; tarpaulin-covered tipper loads; tyre-wash bay at exit gate; CAAQMS[14] (Continuous Air Quality Monitoring Station) at the downwind lease boundary with real-time data to MPCB[15]/CPCB[16]; DG set with acoustic enclosure.

8.2 Noise and Blasting Vibration

Vibration from blasting has been calculated using the USBM RI 8507 Scaled Distance Formula. With charge per delay limited to ≤5 kg and NONEL initiation:

- Predicted Peak Particle Velocity (PPV[17]) at Khanivade village (1.8 km): 3.2 mm/s
- DGMS safe limit for residential structures: 10 mm/s
- PPV is only 32% of the DGMS limit — safe for all structures
- Air overpressure at nearest dwelling: below 133 dB(L) (DGMS limit)

Mitigation: Blasting restricted to 10:00–17:00 hrs only (no night blasting); maximum 20 holes per round; NONEL initiation; muffler mats on boundary blasts; 500 m exclusion zone cleared before every blast; advance blast notice to village Sarpanch and police; engineering seismograph at every blast event; pre-blast structural survey of all buildings within 500 m.

8.3 Water Environment

- No perennial streams within the lease — no direct impact on rivers or water bodies
- Zero discharge policy — all run-off collected in RCC-lined garland drain → 3 settling sumps → reused for dust suppression
- No ground water abstraction —no aquifer dewatering planned
- No acid mine drainage — basalt is geochemically inert with no sulphide minerals
- Domestic sewage from site office treated through bio-toilet + soak pit — no

[14] CAAQMS : Continous Air Quality Monitoring Station
[15] MPCB : Maharashtra Pollution Control Board
[16] CPCB : Central Pollution Control Board
[17] PPV : Peal Paticle Velocity

discharge

8.4 Land and Overburden

- Topsoil stripped separately and conserved on a dedicated 0.30 Ha pad for reuse in plantation and reclamation
- Total OB (9,17,280 T over 5 years) managed in an engineered dump within the lease — toe-wall, garland drain, 1:2 slope, compacted layers
- In-pit backfilling of waste rock commences from Year 3 onwards — reducing surface dump volume
- No dumps outside the lease boundary — no additional land required

8.5 Biological Environment

Impact is assessed as LOW to MODERATE — degraded scrub, no Schedule-I megafauna, no Protected Area within 10 km.

- 1,000+ saplings in peripheral greenbelt (7.5 m wide) + 15,390 saplings total in-lease plantation
- Compensatory Afforestation on 100.19 Ha non-forest land in Parbhani District
- Blasting restricted to 10:00–17:00 hrs only — no disturbance to dawn/dusk wildlife activity
- No night lighting beyond the security perimeter — prevents light disturbance to wildlife
- Annual biodiversity audit of the lease and 5 km buffer, reported in HYCR

8.6 Socio-Economic Impact

The project has NO negative socio-economic impacts as there is no displacement, no land acquisition, and no loss of livelihood (the lease is uninhabited Reserve Forest). Key POSITIVE impacts:

- 9 direct jobs + 25–30 indirect jobs — minimum 60% from Khanivade and adjoining villages
- Royalty to Government of Maharashtra: Rs. 6,222 Lakhs over mine life
- DMF[18] (District Mineral Foundation), Palghar: Rs. 1,867 Lakhs — for district infrastructure
- NMET[19] contribution: Rs. 124 Lakhs
- The Gram Panchayat has proposed a CSR fund of ₹40 lakhs, out of which

[18] DMF : District Mineral Foundation
[19] NMET : National Mineral Explotation Trust

a major portion will be utilized for village development works and the remaining amount will be distributed among the Gram Panchayats.

- CER activities: Rs. 2.06 Lakhs for village road, school, drinking water, health camps, solar lighting
- Approach road (4 km) widened and strengthened by JNPA — permanent community benefit
- Vadhvan Port project: enables 1–1.2 million downstream jobs at full port development

9. ENVIRONMENTAL MANAGEMENT PLAN (EMP)

A comprehensive EMP has been prepared covering all environmental components. Total EMP outlay is Rs. 1.03 Crore over the project life approximately 19% of project capital cost.

Table 15 Environmental Management Plan (EMP Cost)

Sr.	EMP Activity	Capital Cost	Recurring Cost / Year
1	Garland drains & storm water management	5,00,000	75,000
2	Check bunds / silt traps	2,50,000	—
3	Boundary fencing (critical areas)	5,00,000	—
4	Water tanker & dust suppression system	5,00,000	2,25,000
5	Signage boards	75,000	—
6	First aid facilities & PPE	1,50,000	75,000
7	Tree plantation & greenbelt development	76,95,000	7,70,000
8	Noise control measures	2,00,000	—
9	Site office with first-aid room	5,25,000	—
10	Air & noise monitoring	—	90,000
11	Water quality monitoring	—	60,000
12	Statutory compliance & documentation	—	1,00,000
	Total	**1,03,95,000**	**13,95,000**

9.1 Corporate Environmental Responsibility (CER[20])

Table 16 Corporate Environmental Responsibility Cost:

Activity	Beneficiary	Budget (Rs.)
Village internal road repair and culverts	Khanivade Village	51,500
Solar street lighting (10 nos. of 60W LED)	Khanivade Village	20,600
ZP School upgradation (furniture, library, computers, toilets)	ZP School, Khanivade	30,900
RO drinking water unit + 2 hand-pump rejuvenation	Khanivade Village	51,500
Plantation in community areas	Khanivade & adjoining villages	30,900
Awareness programmes (environment, hygiene)	Schools / community	20,600
TOTAL CER		Rs. 2,06,000

10. RISK ASSESSMENT AND DISASTER MANAGEMENT PLAN

A Risk Assessment using the Risk Assessment Code (RAC[21]) approach has been carried out. All post-mitigation risks are LOW to MEDIUM acceptable for this scale of quarry.

Table 17 Risk Assessment and Disaster Management Plan:

Risk / Hazard	Probability	Severity	Risk Level	Key Control Measures
Bench/slope failure	Low	High	Medium	FoS ≥1.5; monthly inspection; weekly in monsoon
Fly-rock from blasting	Low	High	Medium	Stemming ≥0.6 m; muffler mats; 500 m exclusion zone

[20] CER : Corporate Environmental Responsibility

[21] RAC : Risk Assessment Code

Risk / Hazard	Probability	Severity	Risk Level	Key Control Measures
Blast mis-fire	Low	High	Medium	NONEL initiation; 30-min wait SOP; blaster's licence
PPV vibration to structures	Low	Low	Low	Charge/delay ≤5 kg; PPV ≤3.2 mm/s at village
Mine inundation (monsoon)	Low	Medium	Low	Garland drain; 2×50 HP dewatering pumps; SOP
Heavy vehicle accident	Medium	Medium	Medium	Speed governors; road watering; 06:00–18:00 hrs only
Fire (DG / fuel storage)	Low	Medium	Low	Bunded fuel store; DCP extinguishers; fire drill
Magazine theft / fire	Low	High	Medium	Double lock; armed guard; CCTV; daily reconciliation
Medical emergency	Medium	High	Medium	First-aid (Scale-C); AED; PHC Boisar tie-up (7 km)

10.1 Emergency Response Organisation

- Chief Incident Controller: Mines Manager
- Deputy: Foreman
- Safety Officer blast safety, PPE, first-aid coordination
- Licensed Blaster all explosive emergencies and mis-fire handling
- First-Aid Team trained first-aider with AED and Scale-C first-aid kit

10.2 Emergency Contacts

Table 18 Emergency Contacts

PHC Boisar	5 km from site
Civil Hospital, Palghar	28 km
Boisar Fire Station	8 km
Police Station, Boisar	18 km
DGMS Regional Office, Mumbai	Contact number displayed at site

DDMA, Palghar	Contact number displayed at site
National Ambulance	108

Mock drills will be conducted quarterly in Year 1 and half-yearly from Year 2 onwards. After-action reports will be submitted to DGMS and DDMA Palghar within 7 days of each drill.

11. PROGRESSIVE AND FINAL MINE CLOSURE PLAN

11.1 Progressive Mine Closure Plan(PMCP) — Year by Year

Table 19 Progrssive Mine Closure Plan year by year

Year	Key Closure Activity
Year 1	Topsoil stripped and conserved; safety zone fenced; garland drain and sumps commissioned; OB dump toe-wall; 200 saplings planted
Year 2	OB dump shaped and grass-seeded; check dams installed; CA Year-1 monitoring; 200 saplings planted
Year 3	In-pit waste rock backfilling commences; upper bench faces stabilised; 200 saplings planted
Year 4	Upper bench re-vegetation with topsoil + saplings; OB dump capped; 200 saplings planted
Year 5	Final bench worked; FMCP initiated; 200 saplings planted; CA Year-4 monitoring

11.2 Final Mine Closure Plan (FMCP) — After Year 5

- Worked-out pit (18 m deep) converted to a perennial water reservoir handed to Gram Panchayat, Khanivade for community irrigation and drinking water use
- OB dump terraced, capped with topsoil and densely planted evolving into a man-made forest matching the surrounding Reserve Forest
- All temporary infrastructure (site office, DG set, magazine, fuel store, equipment) removed within 6 months of cessation
- Approach road transferred to Gram Panchayat / PWD for permanent community use
- Post-closure environmental monitoring (GW quality, plantation survival, slope stability) for 5 years annual report to SEIAA, MPCB, Forest Dept., District Collector

12. OCCUPATIONAL HEALTH AND SAFETY

All mining operations comply with the Mines Act, 1952; Metalliferous Mines Regulations (MMR), 1961; and all DGMS Technical Circulars. Key occupational health hazards and measures:

Table 20 Occupational Health and Safety

Hazard	Worker at Risk	Control Measure
Silicosis (silica dust)	Drill operators, blasters	Wet drilling (mandatory); P3 dust masks; quarterly PFT, chest X-ray
Noise-Induced Hearing Loss	Drill operators, compressor attendants	Earplugs + earmuffs (double protection); quarterly audiometry
Fly-rock injury	All quarry workers	500 m exclusion zone; muffler mats; stemming compliance
Fall from bench height	All workers near bench edge	Portable rail fencing; safety belt; pre-shift face inspection
Explosive handling	Blaster, Mate	Licensed blaster; NONEL system; DGMS misfire SOP
Vehicle accident	Drivers, pedestrian workers	Speed governors; no night operations; PUC certification

12.1 Medical Examination Programme

Table 21 Medical Examination Programme

Examination	Frequency	Coverage
Pre-employment (PFT, audiometry, chest X-ray, vision, BP)	Before joining	All workers
Annual health examination (PFT, audiometry, X-ray, vision)	Every year	All workers
Personal dust dosimetry (silica monitoring in PM10)	Quarterly	Drill operators, blasters
Noise dosimetry	Quarterly	High-noise zone workers
Blood pressure and general health check	Annual	All workers

13. CONCLUSION

Khanivade Stone Quarry Patch-5 is a small but strategically vital quarry that will supply basalt to the Vadhvan Port a project of national importance under the Sagarmala Programme. The following conclusions emerge from the EIA study:

Parameter	Assessment	Conclusion
Air quality	Incremental PM10 at Khanivade: 5.2 µg/m³; Resultant 47 µg/m³ vs NAAQS 100 µg/m³	SAFE — within standards
Blasting vibration	PPV at nearest dwelling: 3.2 mm/s vs DGMS limit 10 mm/s	SAFE — only 32% of limit
Water quality	Zero discharge; no aquifer dewatering; basalt is acid-safe	NO ADVERSE IMPACT
Biological	Degraded scrub RF; no Schedule-I megafauna; nearest PA >18 km	LOW IMPACT
Social	No displacement; no land acquisition; 9 direct + 25 indirect jobs	NET POSITIVE
Forest	Stage-I FC is in process; NPV Rs. 14.51 Cr; CA 100.19 Ha in Parbhani	In process
Plantation	15,390 saplings in-lease + 100.19 Ha CA offsite + 15,116 trees felled compensated	FULLY OFFSET

Based on this comprehensive environmental assessment, Khanivade Stone Quarry Patch-5 is environmentally feasible, ecologically acceptable, socially responsible, and economically justified. JNPA is committed to strictly adhere to the Comprehensive Environmental Framework during the mine life and the 5 year maintenance period after mine closure.

*****END*****